

RUIJIE ZHENG

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🎓 EDUCATION

University of Michigan, Ann Arbor, Engineering college Aug.2025 – Present
Undergraduate student in Computer Science (CS) GPA: 3.925/4.0 (Major: 3.947/4.0)

Shanghai Jiao Tong University (SJTU), Joint Institute Sep.2023 – July.2025
Undergraduate student in Mechanical Engineering (ME) GPA: 3.72/4.0 (48/335)

📖 PUBLICATIONS

- Gordon Tsai, Zewen Sun, Haoran Dai, Haiqi Shen, **Ruijie Zheng**, Haotian Zhang, Yaoyao Qian, Binghui Wang, Kyrie Zhou. “Weather-Guard: A Real-Time Dynamic Contextual Guardrail for Agentic AI Interactions.” *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2027)*. **Under review**.
- Yuxuan Liu, **Ruijie Zheng**, Brandon Kim, Jason Wu, Anhong Guo. “General Accessibility Model: Understanding Accessibility Needs from Computer Use.” *ACM CHI Conference on Human Factors in Computing Systems (CHI 2027)*, Papers. **Under review**.
- Yuxuan Liu, **Ruijie Zheng**, Brandon Kim, Jason Wu, Anhong Guo. “Demonstration of General Accessibility Models: Learning Accessibility Needs from Everyday Computer Use.” *Adjunct Proceedings of the ACM Symposium on User Interface Software and Technology (UIST 2026)*, Demo. **Accepted**.

👥 PROJECTS & RESEARCH

Research Project: General Accessibility Model (GAM) May.2026 – Present
Prof. Anhong Guo, Human-AI Lab University of Michigan, Ann Arbor

- Second author of GAM (UIST’26 demo accepted; CHI’27 paper under review), which learns users’ accessibility needs by observing everyday computer use instead of explicit calibration tasks
- Built the macOS multimodal capture pipeline (screen, assistive-technology audio, input, DOM) and the LLM reflection loop yielding confidence-weighted barrier propositions, applied to accessibility reports, feature recommendation, and just-in-time Web repair

Research Project: Adaptive-Depth Latent Reasoning Models Mar.2026 – Present
Prof. Yan Chen, LIST Lab Northwestern University, Evanston

- Studying how prompt hardness determines the latent reasoning depth a model needs, toward adaptive rather than fixed recurrence
- Probing hidden-state convergence across latent passes to build a halting mechanism, evaluated on accuracy-versus-compute against fixed-depth baselines

Course Project: LLM-Driven Tennis Coaching System Sep.2025 – Dec.2025
University of Michigan, Ann Arbor Computer Vision

- Developed an end-to-end automated pipeline extracting pose landmarks from tennis videos using MediaPipe and Sport2D
- Designed a heuristic keyframe selection algorithm and a phase-aware LLM prompting strategy to synthesize personalized natural-language coaching feedback

Research Project: Sparse Weight-Decomposed LoRA (SDoRA) Aug.2025 – Dec.2025
Prof. Samet Oymak, SOTA Lab University of Michigan, Ann Arbor

- Proposed SDoRA, a unified Parameter-Efficient Fine-Tuning (PEFT) framework combining weight decomposition with learnable sparse gating

- Implemented the SoRA module and trained Qwen2.5-3B models to analyze sparsity dynamics and layer-wise pruning patterns
- Conducted extensive experiments on commonsense reasoning benchmarks (BoolQ, PIQA, ARC-c) to evaluate training stability and performance

Research Program: Water Jet Propulsion Based on Negative Pressure Energy Storage

Nov.2024 – Present

Prof. Bintang Yang, School of Mechanical Engineering Shanghai Jiao Tong University

- Construct the algorithm logic using Simulink and optimize the control through reinforcement learning
- Modeling the spacecraft shell using SolidWorks

Research Program: Multi-Modality Federated Learning

Jan.2024 – Oct.2024

Prof. Qinya Li, School of Computer Science Shanghai Jiao Tong University

- Used PyTorch to reproduce algorithms in federated learning, such as FedAvg and Harmony
- Improved the efficiency of the algorithm in processing multi-modal data

Innovation Program: Cross-Twin-Rotor Unmanned Helicopter

Feb.2024 – Oct.2024

Prof. Wei Xie, Department of Automation Shanghai Jiao Tong University

- Used quaternions to implement the control algorithm of the aircraft
- Assembled part of the aircraft

Course Design: Mars rover model

May.2024 – Aug.2024

Joint Institute, SJTU Mechanical Design

- Took ENGR1000J Introduction to Engineering and was responsible for the 3D modeling of the Mars rover
- Assembled the rover and implemented part of the control algorithm

🚩 LEADERSHIP

Shanghai Jiao Tong University, Joint Institute

Sep.2024 – Present

Vice President of the American Society of Mechanical Engineers (ASME) Student Branch

- Held workshops and lectures on mechanical engineering topics for students in Joint Institute
- Maintain the ASME official webpage in Joint Institute

Shanghai Jiao Tong University, Joint Institute

Sep.2024 – Jan.2025

Teaching Assistant for MATH1860J Honors Mathematics

- Held recitation class and office hours, prepared worksheets for students to review
- Graded homework and exams, held review sessions before exams

Shanghai Jiao Tong University

May.2024 and Aug.2024

Volunteer teaching activity Yingshang, Anhui

- Deliver lectures on basic physics and mathematics for primary school students in rural areas
- Organize extracurricular activities for students

⚙️ SKILLS

- **Software:** MATLAB, SolidWorks, Mathematica, PowerPoint
- **Design & Prototyping:** Arduino, Raspberry Pi, 3D printing, CO2 laser cutting